MIL-I-8974A
15 July 1981
SUPERSEDING
MIL-I-8974
28 October 1968

MILITARY SPECIFICATION

INDICATOR, EVENT COUNTING, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification covers the general requirements for hermetically sealed indicators, event counting, 28 volts dc, 26 volts ac, 400 Hz, 115 volts ac, 400 Hz or 60 Hz

2 APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel, Plating (Electrodeposited)
QQ-Z-325	Zinc, Coating, Electrodeposited, Requirements
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-676	Boxes Setup

MILITARY

MIL-P-116	Preservation-packaging, Methods Of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5541	Chemical Films, and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

beneficial comments (additions, deletions, recommendations) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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MILITARY (continued)

MIL-I-8974/1	Indicator, Events, Miniature, Digital, 28 VDC
MIL-I-8974/2	Indicator, Events, Miniature, Digital, 28 VDC
MIL-T-10727	Tin Plating, Electrodeposited or Hot-Dipped for Ferrous and Nonferrous Metals

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electric Component Parts
MIL-STD-462	Electromagnetic Interference, Test Requirements and Test Methods
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-889	Dissimilar Metals
MIL-STD-1285	Marking of Electric and Electronic Parts
MS3345	Flange, Split Mounting

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3 REQUIREMENTS

3.1 Specification sheets The individual indicator requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Qualification The indicator furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for opening of bids (See 4.3 and 6.4).

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3.3 Selection of government documents Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3 3 1

3.3.1 Standard parts Standard parts (AN and MS) shall be used wherever they are suitable for the purpose, and shall be identified on any drawings by their part numbers Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used provided they posess suitable properties, are replacable by the standard parts (AN and MS) without alteration, and provided the corresponding standard part numbers are referenced in the parts list, and if practicable, on the contractor's drawings In the event there is no suitable corresponding standard part in effect on the date of invitation for bids, commercial parts may be used provided they conform to all the requirements of this specification

3 4 Materials Materials shall conform to applicable specifications and shall be as specified herein Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended

3 4 1 Critical materials Non-critical materials shall be used where practicable Where the use of a critical material is essential to meet specification requirements, the material used shall be of the least critical of those which are adequate for the purpose

3.4.2 Non-magnetic materials Non-magnetic materials shall be used for all parts except where magnetic materials are essential

3 4 3 Dissimilar metals Dissimilar metals as defined in MIL-STD-889, shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings

3 4 4 Aluminum-alloy parts Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except parts contained in a hermetically sealed enclosure or when any of the following conditions prevail

(a) Small holes and case inserts need not be anodized

(b) Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541

(c) Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541, in lieu of anodizing shall be used

(d) Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

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(e) When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.4.5 Iron and steel parts. Iron and steel parts shall be chromium nickel or zinc plated in accordance with QQ-C-320, QQ-N-290 or QQ-Z-325, respectively. Parts of corrosive resistant steel need not be plated.

3.5 Vacuum impregnation. Coils need not be impregnated when specified performance can be met without impregnation.

3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification and the applicable specification sheet. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.7 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent as approved by the procuring activity.

3.8 Design and construction. The indicator shall be in accordance with the applicable specification sheet and the requirements specified herein. In the event of any conflict between the requirements of this specification and the applicable specification sheet, the latter shall govern. The indicator shall be constructed so that no part will work loose in service. It shall be built to withstand the strains, jars, vibration and other conditions incident to shipping, storage, installation and service.

3.8.1 Counting mechanism. The indicator counting mechanism shall have a digital readout, with a maximum count of 9999 events. The faceplate shall have the word "EVENTS" or a dull black finish. The numerals and scale shall be white on a dull black background. The contractors name or logo may appear on the faceplate.

3.8.2 Enclosure. The case shall be in accordance with the detail specifications and shall provide a hermetically sealed enclosure for the internal mechanism of the instrument. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.8.2.1 Mounting flange. Indicators furnished with a mounting flange shall have the flange permanently attached to the enclosure. The flange shall have the front surface of the mounting flange tin plated in accordance with MIL-T-10727 to provide the case with an electrical ground.

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3.8.2 1.1 Split mounting flange. Indicators provided without mounting flanges may utilize a split mounting flange in accordance with MS3345.

3.8.2.2 Finish. The case shall be finished in durable dull black color

3.8.3. Filling medium The filling medium shall be of at least 98 percent purity and free of dust particles, it must contain not more than 0 006 milligrams of water vapor per liter (dew point -85oF) (-65oC) at the filling pressure, it shall consist of 100 percent helium preferred or a mixture of 88 to 92 percent nitrogen and the remainder helium The absolute pressure of the filling medium in the case shall be a minimum of 1 atmosphere

3 8 4 Coils The coils shall be properly insulated from the case and other grounded parts

3.8 4.1 Coil temperature. The coil temperature rise shall be left to the discretion of the contractor, but shall be considered additive to the maximum operating temperature specified herein.

3.8 5 Terminals All terminals used for soldered connections shall be of the "hook-type" and shall be hot-tin dipped in accordance with MIL-T-10727 or solder coated

3.8 6 Operating position The indicator shall be designed to operate in any position

3 8 7 Interchangeability All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000

3.9 Performance

3 9 1 Operational check The 115 volt 60 Hz and 400 Hz indicators shall operate over the voltage range of 102 to 128 volts AC. The 60 Hz AC indicator shall be capable of operating over the frequency range of 55 to 65 Hz and the 400 Hz indicator shall be capable of operating over the frequency range of 380 to 420 Hz The 28 volt DC type shall operate on a direct current potential ranging from 23 to 29 volts (See 4.6.2)

3 9.2 Power supply tolerance The indicator shall be of such design that it will not be damaged when subjected to power supply frequencies other than specified limits for a period of 30 minutes in any 4-hour period, and shall automatically resume normal operation when the frequency returns within limits, however, there are no requirements on indicator performance at frequencies other than the established limits of 380 to 420 Hz for the 400 Hz ac type. Down to 300 Hz, the voltage shall be considered as maintained within the operating limits The maximum frequency to which the indicator may be subjected without damage shall be 500 Hz

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For the 60 Hz indicator the voltage shall be considered as maintained within the operating limits down to 50 Hz The maximum frequency to which the indicator may be subjected without damage shall be 70 Hz. The 28 volt dc type shall be capable of operating on a dc potential of 23 volts for 30 minutes and 29 volts for 30 minutes without damage (see 4 6 3)

3 9.3 Insulation resistance Unless otherwise specified, the insulation resistance between terminals and case shall exceed 100 megohms (See 4 6 4)

3.9 4 Dielectric stress The indicator will be capable of withstanding application for not more than one minute of 500 volts rms, 60 Hz commercial line frequency between the terminals and case while at an external barometric pressure of 0 82 inch of mercury without damage, arcing or breakdown (See 4 6 5)

3 9 5 Altitude The indicator shall operate within barometric pressure ranging from 30 inches of mercury (sea level) down to 0 82 inch of mercury (80,000 feet elevation) and with pressure varying at a minimum rate of 0 5 inch of mercury per second (See 4 6 6)

3 9 6 Temperature cycling The indicator shall be designed to operate continuously or intermittently, through temperature ranges of -55°C to -85°C, -55°C to 125°C, or -65°C to 218°C ambient, as specified on the applicable specification sheet It must remain within the appropriate temperature range throughout the indicators designed life (See 4 6 7)

3 9 7 Moisture resistance The indicator shall meet the requirements of 4 6 2 after being subjected to the tests specified under 4 6, Test Methods There shall be no signs of moisture in the face of the indicator, nor signs of corrosion (See 4 6 8)

3 9.8 Salt spray The indicator shall meet the requirements of 4 6.2 after being subjected to the tests specified under 4 6, Test methods There shall be no signs of corrosion of metal parts after the test (See 4 6 9)

3 9 9 Vibration The indicator shall operate during and after subjection to the vibration tests as specified in Section 4 Application of the vibration shall be in each of the three principle axes (see 4 6 10)

3 9 10 Shock The indicator shall operate after subjection to the shock tests as specified in section 4 (See 4 6 11)

3 9 11 Hermetic seal The unit shall operate satisfactorily after being subjected to the tests specified in Section 4 The indicator face shall show no evidence of the presence of moisture (See 4.6.12).

3.9.12 Minimum life The indicator shall have a minimum life of 1,000,000 counts (See 4 6 13)

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3.9.12.1 Expected life. The indicator shall be capable of satisfying the minimum life requirement any time within two years (See 4.6.13)

3 9 13 Electromagnetic compatibility. The indicator shall be electromagnetically compatible (See 4 6 14)

3.9 14 Terminal stress Each terminal shall withstand a 4.5 pound (2 04 kg) pull without loosening or being damaged (See 4 6 15)

3.9.15 Storage. The indicator shall be able to withstand storage over extended periods of time at -80°C (See 4 6 16)

3 10 Identification of product Indicators and assemblies and parts thereof shall be marked for identification in accordance with MIL-STD-130 (See 5.2)

3 10 1 Nameplate data The following nameplate data shall appear on the outer su of the enclosure

Manufacturer's name or logo
Manufacturer's model number
Nominal voltage and frequency (if applicable)
Nominal power (watts)
Military number
Coded date (Month/year) per MIL-STD-1285

3 10.2 Use of AN or MS designators AN or MS designations shall not be applied to a product, except for qualification test samples, nor referenced to in correspondence, until notice of approval has been received from the activity responsible for qualification

3 11 Riveting Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed

3 12 Cleaning The indicator shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly

3 13 Workmanship The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, accuracy of dimensions, radii of fillets and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring and riveting, alignment of parts and tightness of assembly, screws and bolts etc

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may use his own facilities or any commercial facility acceptable to the Government The Government reserves the right to perform any of the inspections set forth in this specification when such inspections are necessary to assure that supplies and services conform to the prescribed requirements

4.2 Classification of inspection The examination and testing of indicators shall be as classified below

a Qualification inspection Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products (see 3 2 and 4 3)

b Quality conformance inspection Quality conformance inspection consist of examinations and tests performed on individual products to determine conformance of the products with the requirements set forth in this specification (See 4 4)

4 3 Qualification

4 3 1 Qualification samples The qualification samples shall be plainly identified by durable tags securely attached, and marked with the following information·

Sample for Qualification Inspection
Indicator, Event Counting
Submitted by (Manufacturer's name, date) for
Qualification Inspection in accordance with
MIL-I-8974 under contract or order number
Manufacturer's part number

4 3 2 Qualification inspection Qualification inspection shall consist of the examinations and tests specified in Table I The qualification test samples shall consist of three indicators Failure of any indicator to pass all qualification tests shall constitute failure of all three indicators in the test sample The indicators shall be returned to the supplier to be evaluated as to cause of failure Prior to resubmittal of the test samples, the supplier shall satisfactorily establish that corrective action has been taken

4 3 3 Retention The retention of qualification shall consist of the annual life test and periodic verification to determine compliance of the qualified indicator with the requirements of this specification Periodic verification shall be by certification unless otherwise specified by the activity responsible for the Qualified Products List and shall be at intervals of not more than two years

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TABLE I. Qualification Inspection

Tests	Paragraph	
	Requirement	Method
Examination of product	3 8 3 8 1 3.8.3 3.8.3 1 3 8.3 2 3 8.6 3.6	Visual inspection 4 6 1
Operational check	3 9.1	4 6.2
Power supply tolerance	3 9.2	4 6 3
Insulation resistance	3 9 3	4.6 4
Dielectric stress	3 9 4	4 6 5
Altitude	3 9.5	4 6 6
Temperature cycling	3 9 6	4.6 7
Moisture resistance	3 9 7	4 6.8
Salt spray	3 9 8	4 6 9
Vibration	3.9.9	4 6.10
Shock	3 9 10	4 6 11
Hermetic seal	3 9 11	4 6 12
Life	3 9 12	4 6 13
Electromagnetic compatability	3 9.13	4.6 14
Terminal stress	3 9 14	4 6 15
Low temperature storage	3 9 15	4 6 16

4 4 Quality conformance inspection The quality conformance inspection shall consist of the following tests

(a) Individual tests (4 4 1)
(b) Sampling plans and tests (4 4.2)

4 4.1 Individual tests Each indicator shall be subjected to the following examinations and tests

(a) Examination of product (4.6.1)
(b) Operational check (4 6 2)

4 4 2 Sampling plans The Sampling plans shall consist of Sampling Plan A and Sampling Plan B tests The test samples selected for sampling tests shall first have passed the individual tests The test samples which have been subjected to Sampling Plan A tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling Plan B tests shall not be delivered on contract

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Quantity Offerred for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (Zero when Sampling Plan B is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

When a defective indicator is detected, no indicators from those still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and appropriately corrected In addition, when a failure occurs, shift to one sample out of the next fifteen and proceed as indicated (one from next 50, etc)

4 4 2.1 Scope of Sampling Plan A Test Each sample selected for Sampling Plan A tests shall be subjected to the following tests in the order listed

(a)	Examination of product	(4 6 1)
(b)	Operational check	(4 6 2)
(c)	Power supply tolerance	(4 6 3)
(d)	Temperature cycling	(4 6 [illegible])
(e)	Hermetic seal	(4 6.2)

4 4 2 2 Sampling Plan B tests One indicator shall be selected at random from the first fifteen produced on contract and submitted within ten days after manufacture and thereafter one out of every thousand indicators or part thereof on contract shall be submitted for testing These samples shall be forwarded at the contractor's expense to a government laboratory designated by the procuring activity Each sample shall be plainly identified by a durable tag securely attached and marked with the following information

Indicator, Event Counting, Electromechanical
Submitted by (Manufacturer's name, date)
for production Acceptance Sampling Plan B
Tests of MIL-I-8974 in accordance with
Contract/Order No _____
Manufacturer's Part Number _____

4 4 2 2 1 Sampling Plan B Approval Approval of Sampling Plan B indicators shall be by the procuring activity upon satisfactory completion of the designated tests Any design, material or performance defect made evident during these tests shall be corrected by the contractor to the satisfaction of the procuring activity Failure of the sample units to pass any of the tests shall be cause for the deliveries of indicators under the contract to cease until proper corrective action is approved and accomplished

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4 4 2 2 2 Scope of Sampling Plan Test B Each sample selected for Sampling B tests shall be subjected to the following tests in the order listed

(a)	Examination of product	(4 6 1)
(b)	Operational check	(4 6 2)
(c)	Power supply tolerance	(4 6 3)
(d)	Temperature cycling	(4 6 7)
(e)	Salt spray	(4 6 9)
(f)	Vibration	(4 6 10)
(g)	Shock	(4 6 11)
(h)	Hermetic seal	(4 6 12)

4 4 2 3 Rejection and retest When one item selected from a production run fails to meet the specification requirements items still on hand or later produced will not be accepted until the extent and cause of the failure are determined After corrections have been made, all necessary tests shall be repeated Individual tests may continue for operational and production reasons, pending the results of the investigation of a sampling test failure Final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification

4 4 3 Defects in items already accepted The investigation of a test failure could indicate that defects may exist in items already accepted If so the contractor shall fully advise the procuring activity of all defects likely to be found, and methods for correcting them

4 5 Test conditions Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-202.

4 5 1 Standard atmospheric conditions Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29 92 inches of mercury) and at room temperature (approximately 25oC) When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition

4 5 1 1 Conditions for vibration and shock tests When subjected to vibration and shock tests, the indicator without mounting flanges shall be firmly held in a block or clamp of low resonance aluminum alloy (61s preferred) The whole mass shall then be rigidly attached to the table of the test apparatus

4 6 Test methods

4.6 1 Examination of product The indicator shall be examined visually for conformance to dimensions, workmanship, and other requirements as specified in Table I

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4 6 2 Operational check Operating power shall be applied and removed at the terminals at a rate of 5 pulses per second with a pulse duration of 0 05 +0 005 second Application and removal of operating power shall be maintained for 5 minutes for each of the following conditions

28 volt d c indicator	400 Hz ac indicator	60 Hz ac indicator
23 volts d c	115 volts rms, 400 Hz	115 volts rms, 60Hz
29 volts d c	102 volts rms, 420 Hz	102 volts rms, 65Hz
	128 volts rms, 380 Hz	128 volts rms, 55Hz
	26 volts rms, 400 Hz	
	23 volts rms, 420 Hz	
	29 volts rms, 380 Hz	

At the completion of the test, the accrued counts shall be verified against the number of signals applied The units must be within one count of true reading when compared with a suitable monitoring counter (see 3 9 1)

4 6 3 Power supply tolerance The indicator shall be subjected to steady energization periods for 30 minutes at each of the following conditions

28V dc indicator	400 Hz ac indicator	60 Hz ac indicator
23 volts dc	115 volts rms, 300 Hz	115 volts rms 50 Hz
29 volts dc	129 volts rms, 500 Hz	128 volts rms, 70 Hz
	26 volts rms, 300 Hz	
	29 volts rms, 500 Hz	

Upon completion of the power supply tolerance test, the indicators shall be subjected to the test specified in 4 6 2

4 6 4 Insulation resistance The insulation resistance between terminals and case shall exceed 100 megohms when subjected to Method 302, Condition B of MIL-STD-202 (see 3 9 3)

4 6 5 Dielectric stress The indicator shall be subjected to Method 301 of MIL-STD-202, except that a potential of 500 volts rms, 60 Hz shall be applied at a barometric pressure of 0.82 inch of mercury (corresponding to 80,000 feet altitude) (See 3 9 4)

4 6 6 Altitude The indicator shall be subjected to Method 105 of MILSTD-202, except that the absolute pressure shall be 0 82 inch of mercury (corresponding to 80,000 feet altitude) The units shall be operated at a rate of 5 pulses per second for a period of 1 minute The dc indicator shall be operated at 29 volts dc, the 400 Hz unit shall be operated at 128 volts, 420 Hz, or 29 volts, 420 Hz, as applicable, and the 60 Hz unit shall be operated at 128 volts, 65 Hz (See 3 9.5.).

4 6 7 Temperature cycling The indicator shall be subjected to Method 102, Condition "C" or "D" in accordance with MIL-STD-202 unless other-

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wise specified in the applicable specification sheet The unit shall be operated during step 3 of the test The dc indicator shall be operated at 29 volts, dc, the 400 Hz unit shall be operated at 128 volts, 420 Hz or 29 volts, 420 Hz as applicable, and the 60 Hz unit shall be operated at 128 volts, 65 Hz (See 3 9 6)

4.6.8 Moisture resistance The unit shall be subjected to Method 106, figure 1 titled "Graphic representation of moisture-resistance test" of MIL-STD-202 No voltage shall be applied during this test The indicator shall be mounted with the digital display in a vertical position (normal reading altitude) The indicator shall be subjected to the operational check (4 6 2) before starting and after completion of the moisture resistance test (See 3.9.7)

4 6 9 Salt spray The unit shall be tested in accordance with Method 101, Condition B of MIL-STD-202 (See 3 9 8)

4 6 10 Vibration The indicator shall be tested in accordance with Method 204, Condition D, except 10g maximum vibration Application of vibration shall be in each of the three principal planes The units shall be energized at a rate of five pulses per second for at least ten percent of the vibration in each of the three planes The indicator shall be subjected to the operational check (4 6 2) before starting and after completion of the vibration test (See 3 9.9)

4 6 11 Shock The indicator shall be tested in accordance with MIL-STD-202, Method 213, Condition A, using test equipment per Method 202, to an acceleration of 50g for 11±1 millisecond (See 3 9 10)

4.6 12 Hermetic seal Indicators having a filling medium of helium or helium-nitrogen mixture shall be tested for leaks be means of a mass spectrometer type of helium leak detector The leak rate for the helium-nitrogen mixture shall not exceed 0 01 micron cubic foot per hour Where 100 percent helium is utilized as a filling medium, the leak rate shall not exceed 0 1 micron cubic foot per hour As an alternate method, the helium (or nitrogen-helium) filled units may be subjected to Method 104 condition A of MIL-STD-202 (See 3 9 11)

4 6 13 Life As part of the qualification test procedure, three samples of each type shall be placed on life test and operated at a rate of 5 pulses per second for a total of 1,000,000 pulse cycles If any one of the indicators fails, a new group shall be used to repeat the entire test until a group of three has passed This test shall be repeated at least once a year following preconditioning tests of temperature cycling, shock and vibration Failure to meet the annual life test shall be cause for removal of the indicator from the qualified products list (See 3 9 12 and 3.9.12 1)

4.6 14 Electromagnetic compatibility. Radiated and conducted interference control tests shall be conducted in accordance with MIL-STD-462 (See 3 9 13)

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4.6 15 Terminal stress. Each terminal shall be tested in accordance with MIL-STD-202, Method 211, test condition "A" (See 3 9.14.).

4 6 16 Low temperature storage The indicator shall be placed in a chamber maintained at -60°C, humidity normal for temperature for one half hour The temperature of the indicator shall then be raised to -50°C +5°C and the indicator operationally checked (4 6 2) This procedure shall be repeated three times At the completion of this test the indicator shall be brought to standard atmospheric conditions (4 5 1) and conform to the requirements of 4.4 1 (See 3 9.15)

4 6.17 Packaging inspection The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with MIL-STD-[illegible] (See 5 2)

5 PRESERVATION - PACKAGING

5 1 Preservation, - packaging Preservation - packaging shall be in accorcance with level A or C, as specified (see 6 2)

5 1 1 Level A

5 1 1 1 Cleaning Indicators shall be cleaned in accordance with MIL-P-116 process C-1

5 1 1.2 Drying Indicators shall be dried in accordance with MIL-P-116

5 1 1 3 Preservatives Preservatives shall not be used

5 1 1 4 Wrapping and cushioning Materials used shall be nonstatic generating, noncorrosive and shall not crumble, flake powder or shed or be of fibrous construction

5 1 1 5 Unit packaging Unless otherwise specified, indicators shall be individually packaged in accordance with submethod 1C-3 of MIL-P-116, as a minimum ½ inch of cushioning shall be provided around each surface of the indicator with special emphasis on the connectors and read-out window

5 1 1 6 Intermediate packaging Indicators packaged as specified in 5 1 1 5 shall be placed in intermediate containers conforming to PPP-B-676 or PPP-B-636 Intermediate containers shall be uniform in size, shape and quantities, shall be ofminimum tare and cube and shall contain multiples of five unit packages

5 1.2 Level C. Indicators shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

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5 2 Packing. Packing shall be level A, B, or C as specified (see 6.2) in accordance with the requirements of MIL-STD-794.

5 3 Marking In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 The use of labels for identification marking on exterior containers is not permitted.

5.3 1 Reinspection dates. Reinspection marking will be as specified by the procuring activity. Unpacked items not in a shipping container shall have firmly affixed thereto a tag or label with essentially the same legend

6 NOTES

6 1 Intended use The indicators, event counting, covered by this specification are intended for use in equipment where there is a requirement for accurately recording the number of times the equipment has been energized

6 2 Ordering data Procurement documents should specify the following

(a) Title, number and date of this specification
(b) Part number of the detail specification sheet
(c) Level of preservation, packaging packing and marking (see 5 1 and 5 2)
(d) Reinspection dates (see 5.2 1)
(e) Requirement for Sampling Plan A and/or Sampling Plan B (See 4 4.2)

6 3 Data For the information of contractors and contracting officers any of the data specified in (a) applicable documents listed in Section 2 of this specification or (b) referenced lower tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirement List) which shall be attached to and made part of the contract or order For Navy contracts, NavWep Form 4200/15 Drawings, Lists, and Specifications Required shall be attached where applicable.

6 4 Qualification With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date The attention of the suppliers is called to this requirement and manufacturer's are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification The activity responsible for the Qualified Products List is Commander, Naval Air Systems Command, Attention AIR-5162F, Washington, DC 20360, and information pertaining to qualification of products may be obtained from that activity

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6 5 Changes from previous issue Asterisks are not used in this revision to identify changes with respect to the previous issue, due to extensiveness of the changes.

6 6 Definitions

6.6 1 Voltages All alternating current voltages as used in this specification will be considered to be a root mean square (rms) values

6 6.2 Hermetic seal A hermetic seal is defined as a perfectly closed and airtight seal A hermetic seal is not intended to include seal accomplished by gaskets

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